

Provincial Agricultural Drought Mitigation Planning: Optimal LDRRMF Allocation for Cavite Province

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1 Introduction

In municipal climate-adaptation planning, local governments are faced with the challenge of deploying scarce fiscal capital to mitigate severe environmental hazards. In agricultural economies such as the Province of Cavite, prolonged moisture deficits triggered by the El Niño Southern Oscillation (ENSO) threaten crop yields, lower soil water retention, and jeopardize food security. Historically, local disaster resource allocations have relied heavily on uncalibrated intuition often leaving the most ecologically vulnerable crop regions severely underfunded.

To move toward an audit-ready, evidence-based governance paradigm, operational research integrates predictive machine learning with constrained mathematical programming. By combining multi-year satellite climatological data with statutory fiscal boundaries, local disaster councils can formulate resource distribution as a continuous **Linear Programming Problem (LPP)**. The objective is to deploy public resources optimally to maximize risk-weighted disaster protection while guaranteeing legal compliance, municipal operational readiness, and regional agricultural equity.

2 Statement of the Problem

The Provincial Disaster Risk Reduction and Management Council (PDRRMC) of Cavite is preparing its fiscal allocation plan for the peak dry season (February to May). The province comprises twenty-three (23) local government units ($n = 23$), consisting of 16 municipalities and 7 component cities.

A two-stage analytical pipeline has been developed:

1. A Random Forest regression model trained on multi-year NASA POWER agro-meteorological data predicted the mean dry-season root-zone soil wetness ($\overline{\text{GWETROOT}}_i \in [0, 1]$) for each LGU. The empirical drought risk score is defined as the complementary moisture deficit:

$$R_i = 1 - \overline{\text{GWETROOT}}_i, \quad \forall i \in \{1, 2, \dots, 23\}. \quad (1)$$

2. A linear programming model that distributes the pre-disaster preparedness budget (x_i) to maximize total risk-weighted impact subject to statutory public finance laws and physical framework constraints.

The optimization model follows two types of rules: **national legal mandates** (from RA 10121) and **project benchmark parameters** (realistic example values chosen for this case study).

1. **Legal Minimum for Local Calamity Funds (Statutory Law):** Under **Republic Act No. 10121, Section 21, Paragraph 1**, every local government unit (LGU) must set aside at least five percent (5%) of its regular yearly revenue for the Local Disaster Risk Reduction and Management Fund (LDRRMF, denoted as T).

2. **The 30% Quick Response Fund Reserve (Statutory Law):** Under **RA 10121, Section 21, Paragraph 2** and **NDRRMC–DBM–DILG Joint Memorandum Circular No. 2013-1, Section 5.1**, exactly thirty percent (30%) of the LDRRMF must be set aside as a Quick Response Fund (QRF, denoted as $Q = 0.30T$). By law, this emergency money is on stand-by for relief work and cannot be budgeted or spent before an official State of Calamity is declared.

3. **The 70% Pre-Disaster Preparedness Budget (Statutory Law):** Because 30% is locked for emergency relief, the remaining seventy percent (70%) of the fund is used for proactive disaster prevention and preparedness ($B = 0.70T$). The linear programming model allocates only this 70% preparedness budget:

$$\sum_{i=1}^{23} x_i \leq B. \quad (2)$$

4. **Total Provincial Budget Example (Benchmark Value):** While the 5% rule is set by law, the exact peso amount depends on the province’s revenue. For this research project, we use a representative benchmark of:

$$T = \text{PHP } 50,000,000.$$

Following the legal 70/30 split, this sets the preparedness budget ceiling at $B = \text{PHP } 35,000,000$, and leaves $Q = \text{PHP } 15,000,000$ in the locked emergency reserve.

5. **Minimum Baseline Allocation (Benchmark Value):** The law does not specify an exact peso floor for each town. To ensure every municipality receives enough money to maintain weather sensors, transport, and local staff, we set a minimum policy floor of $L_i = \text{PHP } 400,000$:

$$x_i \geq 400,000 \quad \forall i \in \{1, \dots, 23\}. \quad (3)$$

6. **Maximum Allocation Cap (Benchmark Value):** To prevent any single city or town from taking too much of the shared budget, we set an administrative cap of fifteen percent (15%) of the preparedness budget ($U_i = 0.15B = \text{PHP } 5,250,000$):

$$x_i \leq 5,250,000 \quad \forall i \in \{1, \dots, 23\}. \quad (4)$$

7. **Upland Farming Cluster Share (Benchmark Value):** According to the *Cavite Provincial Development and Physical Framework Plan (PDPFP)*, the province’s coffee, cacao, and rain-fed crops are located in the nine (9) upland towns:

$$\mathcal{U} = \{\text{Alfonso, Amadeo, Gen. Emilio Aguinaldo, Indang, Magallanes, Maragondon, Mendez, Silang, Tagaytay City}\}$$

Because rain-fed farms face higher drought risk than irrigated lowland areas, we set a policy rule that these 9 towns must collectively receive at least sixty percent (60%) of the total preparedness budget ($0.60B = \text{PHP } 21,000,000$). The remaining balance (up to 40%, or $\text{PHP } 14,000,000$) is available for the 14 lowland towns and cities:

$$\sum_{i \in \mathcal{U}} x_i \geq 0.60B. \quad (5)$$

3 Tasks

As quantitative policy consultants to the Provincial Government of Cavite, you are tasked with completing the following deliverables:

1. **Mathematical Formulation:** Formally define the continuous Linear Programming Problem. Clearly label decision variables, the risk-weighted objective function, and express all policy constraints in standard mathematical inequality notation. Show how the 30% Quick Response Fund (Q) is explicitly partitioned under RA 10121, Section 21.

1. Statutory Fund Partition and Parameter Definitions

In strict adherence to Section 21 of Republic Act No. 10121 and Joint Memorandum Circular No. 2013-1, the benchmark Local Disaster Risk Reduction and Management Fund ($T = \text{PHP } 50,000,000$) is partitioned into two statutory components prior to optimization:

$$Q = 0.30 T = \text{PHP } 15,000,000 \quad (\text{Quick Response Fund, stand-by relief reserve}) \quad (6)$$

$$B = 0.70 T = \text{PHP } 35,000,000 \quad (\text{Statutory Pre-Disaster Preparedness Budget}) \quad (7)$$

Because Q is legally encumbered for post-disaster response during a declared State of Calamity, only the preparedness ceiling B is available for proactive allocation.

Let the operational parameters and regional topology of Cavite be defined as follows:

- $\mathcal{M} = \{1, 2, \dots, 23\}$: The index set denoting all 23 local government units (16 municipalities and 7 component cities) of Cavite ($|\mathcal{M}| = 23$).
- $\mathcal{U} \subset \mathcal{M}$: The subset containing the nine (9) designated upland agricultural LGUs ($|\mathcal{U}| = 9$), leaving 14 lowland LGUs in $\mathcal{M} \setminus \mathcal{U}$.
- $d_i \in \{0, 1\}$: A deterministic parameter encoding regional agricultural zoning:

$$d_i = \begin{cases} 1, & \text{if } i \in \mathcal{U} \\ 0, & \text{if } i \in \mathcal{M} \setminus \mathcal{U} \end{cases}$$

- $R_i \in [0, 1]$: The empirical drought risk score of LGU i , defined by the complement of predicted root-zone soil wetness: $R_i = 1 - \overline{\text{GWETROOT}}_i$.
- $B = \text{PHP } 35,000,000$: The total provincial statutory preparedness budget ceiling.
- $L_i = \text{PHP } 400,000$: The mandatory operational survival floor for each LGU $i \in \mathcal{M}$.
- $U_i = 0.15 B = \text{PHP } 5,250,000$: The individual equity absorption ceiling for each LGU $i \in \mathcal{M}$.

2. Primal Problem Formulation

The resource distribution model is formulated as the following continuous Linear Programming Problem (LPP):

$$\begin{aligned} & \textbf{maximize} \quad Z = \sum_{i \in \mathcal{M}} R_i x_i \\ & \textbf{subject to:} \quad \sum_{i \in \mathcal{M}} x_i \leq B \quad (\text{Statutory Preparedness Budget Ceiling}) \\ & \quad \sum_{i \in \mathcal{M}} d_i x_i \geq 0.60 B \quad (\text{Zonal Upland Agricultural Guarantee}) \quad (8) \\ & \quad x_i \geq L_i, \quad \forall i \in \mathcal{M} \quad (\text{Operational Survival Floor}) \\ & \quad x_i \leq U_i, \quad \forall i \in \mathcal{M} \quad (\text{Equity Absorption Ceiling}) \\ & \quad x_i \geq 0, \quad \forall i \in \mathcal{M} \quad (\text{Non-Negativity}) \end{aligned}$$

where each decision variable $x_i \in \mathbb{R}_{\geq 0}$ denotes the continuous fiscal appropriation granted to municipality or city i .

3. Reduction to Canonical Standard Form

To transform the primal formulation into the canonical standard form ($\min \mathbf{c}^T \mathbf{x}$ subject to $\mathbf{Ax} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$), we perform the following transformations:

- (a) Convert maximization to minimization by negating the objective coefficients: $\max Z \equiv -\min(-Z)$.
- (b) Introduce slack variable $s_1 \geq 0$ for the statutory budget ceiling constraint.
- (c) Introduce surplus variable $e_1 \geq 0$ for the zonal upland guarantee constraint.
- (d) Introduce surplus variables $v_i \geq 0$ for each $i \in \mathcal{M}$ for the operational survival floor constraints.
- (e) Introduce slack variables $u_i \geq 0$ for each $i \in \mathcal{M}$ for the equity absorption ceiling constraints.

The scalar equality system is expressed as:

$$\begin{aligned}
& \text{minimize} && \sum_{i \in \mathcal{M}} -R_i x_i \\
& \text{subject to:} && \sum_{i \in \mathcal{M}} x_i + s_1 = 35,000,000 \\
& && \sum_{i \in \mathcal{M}} d_i x_i - e_1 = 21,000,000 \\
& && x_i - v_i = 400,000, \quad \forall i \in \mathcal{M} \\
& && x_i + u_i = 5,250,000, \quad \forall i \in \mathcal{M} \\
& && x_i, v_i, u_i \geq 0 \quad (\forall i \in \mathcal{M}), \quad s_1, e_1 \geq 0
\end{aligned} \tag{9}$$

4. Matrix Representation

The system comprises $m = 48$ linear equality constraints and $n = 71$ continuous non-negative variables. The augmented solution vector $\mathbf{x} \in \mathbb{R}^{71}$, cost vector $\mathbf{c} \in \mathbb{R}^{71}$, and requirement vector $\mathbf{b} \in \mathbb{R}^{48}$ are partitioned in block form as:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_{23 \times 1} \\ s_1 \\ e_1 \\ \mathbf{v}_{23 \times 1} \\ \mathbf{u}_{23 \times 1} \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} -R_1 \\ \vdots \\ -R_{23} \\ 0 \\ 0 \\ \mathbf{0}_{23 \times 1} \\ \mathbf{0}_{23 \times 1} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 35,000,000 \\ 21,000,000 \\ 400,000 \cdot \mathbf{1}_{23 \times 1} \\ 5,250,000 \cdot \mathbf{1}_{23 \times 1} \end{bmatrix}$$

The structural constraint matrix $\mathbf{A} \in \mathbb{R}^{48 \times 71}$ is assembled as:

$$\mathbf{A} = \begin{bmatrix} \mathbf{1}_{23 \times 1}^T & 1 & 0 & \mathbf{0}_{23 \times 1}^T & \mathbf{0}_{23 \times 1}^T \\ \mathbf{d}_{23 \times 1}^T & 0 & -1 & \mathbf{0}_{23 \times 1}^T & \mathbf{0}_{23 \times 1}^T \\ \mathbf{I}_{23} & \mathbf{0}_{23 \times 1} & \mathbf{0}_{23 \times 1} & -\mathbf{I}_{23} & \mathbf{0}_{23 \times 23} \\ \mathbf{I}_{23} & \mathbf{0}_{23 \times 1} & \mathbf{0}_{23 \times 1} & \mathbf{0}_{23 \times 23} & \mathbf{I}_{23} \end{bmatrix} \tag{10}$$

where $\mathbf{I}_{23}$ is the 23×23 identity matrix, $\mathbf{1}_{23 \times 1}$ is a column vector of ones, and $\mathbf{d} = [d_1, \dots, d_{23}]^T$ is the static upland indicator vector. Because every decision, slack, and surplus variable belongs to $\mathbb{R}_{\geq 0}$, the problem defines a bounded convex polyhedron in $\mathbb{R}^{71}$ solvable via standard continuous Primal Simplex algorithms.

2. **Algorithmic Implementation Formulation:** To execute the optimization numerically via modern solvers such as `scipy.optimize.linprog` (using `method='highs'`), the continuous LPP is cast into the canonical bounded upper-inequality form:

$$\begin{aligned} & \text{minimize} && \mathbf{c}^T \mathbf{x} \\ & \text{subject to} && \mathbf{A}_{\text{ub}} \mathbf{x} \leq \mathbf{b}_{\text{ub}} \\ & && \boldsymbol{\ell} \leq \mathbf{x} \leq \mathbf{u} \end{aligned} \tag{11}$$

Rather than augmenting the system with 46 auxiliary slack and surplus variables for individual municipal floors and ceilings, modern implementations handle box constraints directly via variable bounds. This reduces the structural inequality matrix to only $m = 2$ global coupling constraints over $n = 23$ decision variables:

- **Objective Vector** ($\mathbf{c} \in \mathbb{R}^{23}$): Since `linprog` minimizes by default, the risk maximization objective is converted by setting:

$$\mathbf{c} = -\mathbf{R} = -[R_1, R_2, \dots, R_{23}]^T \tag{12}$$

- **Decoupled Box Bounds** ($\boldsymbol{\ell}, \mathbf{u} \in \mathbb{R}^{23}$): The survival floors and absorption ceilings are isolated from the row constraints and supplied via the `bounds` parameter:

$$\ell_i = L_i = \text{PHP } 400,000, \quad u_i = U_i = \text{PHP } 5,250,000, \quad \forall i \in \mathcal{M} \tag{13}$$

- **Coupling Inequality System** ($\mathbf{A}_{\text{ub}} \in \mathbb{R}^{2 \times 23}$, $\mathbf{b}_{\text{ub}} \in \mathbb{R}^2$): The remaining macro-policy constraints consist of the global statutory budget ceiling and the zonal upland guarantee. Inverting the greater-than inequality of the upland guarantee to conform to the standard $\leq$ format yields:

$$\mathbf{A}_{\text{ub}} = \begin{bmatrix} \mathbf{1}_{23}^T \\ -\mathbf{d}^T \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ -d_1 & -d_2 & \cdots & -d_{23} \end{bmatrix} \in \mathbb{R}^{2 \times 23} \tag{14}$$

and the corresponding right-hand side capacity vector:

$$\mathbf{b}_{\text{ub}} = \begin{bmatrix} B \\ -0.60 B \end{bmatrix} = \begin{bmatrix} 35,000,000 \\ -21,000,000 \end{bmatrix} \in \mathbb{R}^2 \tag{15}$$

This compact formulation matches the mathematical inputs passed directly to the solver routine while eliminating redundant basis inversions.

3. Budget Allocation Analysis

Analysis based on Table 1

- (a) Identify which municipalities reached the maximum absorption ceiling ($U_i = \text{PHP } 5,250,000$) and which remained at the operational baseline floor ($L_i = \text{PHP } 400,000$).
 - **The Maximum Limit (PHP 5,250,000):** Five upland towns (Alfonso, Amadeo, Magallanes, Mendez, and Indang) received the maximum allowed budget. Why? Because our machine learning model identified them as having the highest drought risk (scores around 0.54 to 0.55). We put a "ceiling" (15% maximum) on these towns so that the most at-risk areas get the most help, but they do not drain the entire provincial fund.

- **The Remaining Balance (PHP 1,950,000):** General Emilio Aguinaldo also had a very high risk score (0.528) and was next in line. However, after funding the top five towns and setting aside basic emergency funds for everyone else, the province only had PHP 1,950,000 left. The formula simply gives the exact remaining "change" to this town to completely use up the budget.
 - **The Baseline Minimum (PHP 400,000):** The other 17 towns had lower drought risk scores (0.41 and below). Because their agricultural risk is lower this season, they stay at the "floor" limit. This guarantees that every single town gets a basic PHP 400,000 to keep their local disaster offices running, ensuring no town is left at zero.
- (b) Explain the optimization model's decision-making logic: how did the algorithm balance providing emergency readiness funds to every town, prioritizing the most vulnerable upland farming areas, and preventing any single municipality from receiving an excessive share of public funds?

This is calculated using an algorithm based on George Dantzig's Simplex Method. It is highly reliable because it acts like a strict, logical accountant following a clear set of rules:

- (a) **Pay the Minimums First:** The math instantly sets aside PHP 400,000 for all 23 towns. This locks up PHP 9.2 million of the 35 million budget right away, ensuring basic survival funds for everyone.
- (b) **Rank by Risk:** It takes the remaining PHP 25.8 million and sorts the towns strictly from highest drought risk to lowest.
- (c) **Apply the Cap (Ceiling):** It starts pouring money into the highest-risk towns. Once a town hits the 15% legal limit (PHP 5.25 million), the math cuts them off and moves to the next town on the list.
- (d) **Stop at Zero:** When it reaches the sixth town (General Emilio Aguinaldo), the money runs out. The math stops exactly at the PHP 35 million total.

By using this mathematical method, we can prove to the provincial government that every single peso was assigned based purely on climate data and budget laws, completely removing human bias and politics from the decision.

4. Scenario Analysis: Climate Stress-Testing under an Intensifying El Niño Advisory

Grounded in the official DOST-PAGASA Climate Advisory (Press Release No. 225) warning of an intensifying "Very Strong" El Niño event, conduct a counterfactual sensitivity experiment to evaluate the resilience of the combined machine learning and linear programming allocation pipeline. Rather than manually altering budgetary quotas, apply an empirical climatological perturbation directly to the dry-season evaluation features (February–May):

$$\text{PRECTOTCORR}' = 0.70 \times \text{PRECTOTCORR}, \quad \text{T2M}' = \text{T2M} + 1.5^\circ\text{C} \quad (16)$$

representing a 30% reduction in precipitation (consistent with PAGASA's official dry spell / below-normal criteria) coupled with a +1.5°C surface air temperature anomaly.

Table 1: Optimal LDRRMF Preparedness Allocation via Simplex Method

Municipality	Zone	Risk Score (R_i)	Allocated (PHP)	% of Budget
Ceiling Allocations ($U_i = \text{PHP } 5,250,000$)				
Alfonso	Upland	0.549991	5,250,000	15.00
Amadeo	Upland	0.549991	5,250,000	15.00
Magallanes	Upland	0.549560	5,250,000	15.00
Mendez	Upland	0.549311	5,250,000	15.00
Indang	Upland	0.539777	5,250,000	15.00
Marginal Budget-Clearing Allocation				
Gen. Emilio Aguinaldo	Upland	0.528011	1,950,000	5.57
Baseline Floor Allocations ($L_i = \text{PHP } 400,000$)				
Tagaytay City	Upland	0.407163	400,000	1.14
Silang	Upland	0.388931	400,000	1.14
Maragondon	Upland	0.410000	400,000	1.14
Ternate	Lowland	0.410000	400,000	1.14
Tanza	Lowland	0.410000	400,000	1.14
Rosario	Lowland	0.410000	400,000	1.14
Noveleta	Lowland	0.410000	400,000	1.14
Naic	Lowland	0.410000	400,000	1.14
Kawit	Lowland	0.407000	400,000	1.14
Imus City	Lowland	0.409000	400,000	1.14
General Trias	Lowland	0.410000	400,000	1.14
Gen. Mariano Alvarez	Lowland	0.354754	400,000	1.14
Dasmaringas City	Lowland	0.410000	400,000	1.14
Cavite City	Lowland	0.410000	400,000	1.14
Carmona	Lowland	0.380971	400,000	1.14
Bacoar City	Lowland	0.400923	400,000	1.14
Trece Martires City	Lowland	0.410000	400,000	1.14
Total Budget Distributed			35,000,000	100.00